Why the EM Log-Likelihood Does Not Decrease

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Let

$$\ell(\theta) = \log P(X \mid \theta)$$

be the observed-data log-likelihood.
At iteration t, define

$$q(Z) = P(Z \mid X, \theta^{(t)}).$$

Also recall that

$$Q(\theta, \theta^{(t)}) = E_{Z|X,\theta^{(t)}} \left[\log P(X, Z \mid \theta)\right].$$

Step 1: Rewrite the observed-data likelihood

Since

$$P(X \mid \theta) = \sum_Z P(X, Z \mid \theta),$$

we can multiply and divide by $q(Z)$:

$$P(X \mid \theta) = \sum_Z q(Z) \frac{P(X, Z \mid \theta)}{q(Z)}.$$

Therefore,

$$\ell(\theta) = \log \left[\sum_Z q(Z) \frac{P(X, Z \mid \theta)}{q(Z)}\right].$$

Step 2: Apply Jensen's inequality

Since the logarithm is a concave function, Jensen's inequality tells us that

$$\log E[Y] \geq E[\log Y].$$

Applying this with respect to the distribution $q(Z)$ gives

$$\ell(\theta) \geq \sum_Z q(Z) \log \left[\frac{P(X, Z \mid \theta)}{q(Z)}\right].$$

Expanding the logarithm,

$$\ell(\theta) \geq \sum_Z q(Z) \log P(X, Z \mid \theta) - \sum_Z q(Z) \log q(Z).$$

The first term is exactly the Q-function:

$$\sum_Z q(Z) \log P(X, Z \mid \theta) = Q(\theta, \theta^{(t)}).$$

Hence,

$$\ell(\theta) \geq Q(\theta, \theta^{(t)}) - \sum_Z q(Z) \log q(Z).$$

During the M-step, $q(Z)$ is fixed, so

$$-\sum_Z q(Z) \log q(Z)$$

is a constant with respect to θ. Therefore we can write

$$\ell(\theta) \geq Q(\theta, \theta^{(t)}) + C,$$

where

$$C = -\sum_Z q(Z) \log q(Z).$$

Step 3: The bound is exact at the current parameter value

At $\theta = \theta^{(t)}$,

$$q(Z) = P(Z \mid X, \theta^{(t)}).$$

Using Bayes' rule,

$$P(Z \mid X, \theta^{(t)}) = \frac{P(X, Z \mid \theta^{(t)})}{P(X \mid \theta^{(t)})}.$$

Therefore,

$$\frac{P(X, Z \mid \theta^{(t)})}{q(Z)} = P(X \mid \theta^{(t)}),$$

which does not depend on Z.

Hence the quantity inside the logarithm is constant in Z, so equality holds in Jensen's inequality:

$$\ell(\theta^{(t)}) = Q(\theta^{(t)}, \theta^{(t)}) + C.$$

Step 4: The M-step increases the lower bound

The M-step chooses

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta, \theta^{(t)}).$$

Therefore,

$$Q(\theta^{(t+1)}, \theta^{(t)}) \geq Q(\theta^{(t)}, \theta^{(t)}).$$

Since

$$\ell(\theta^{(t+1)}) \geq Q(\theta^{(t+1)}, \theta^{(t)}) + C,$$

we obtain

$$\ell(\theta^{(t+1)}) \geq Q(\theta^{(t+1)}, \theta^{(t)}) + C$$

and therefore

$$\ell(\theta^{(t+1)}) \geq Q(\theta^{(t)}, \theta^{(t)}) + C.$$

But

$$Q(\theta^{(t)}, \theta^{(t)}) + C = \ell(\theta^{(t)}).$$

Hence,

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})}$$

so the observed-data log-likelihood does not decrease at each EM iteration.

Intuition

The E-step constructs a lower bound to the observed-data log-likelihood that touches the log-likelihood at the current parameter value.

The M-step increases that lower bound.

Because the new observed-data log-likelihood must lie above the new lower bound, it cannot be smaller than the previous observed-data log-likelihood.